DESIGN AND DEVELOPMENT OF A NIGERIAN PUBLIC ECONOMIC DATA AGGREGATION AND ANALYTICS PLATFORM WITH OPEN API ACCESS: A 2020–2026 CASE STUDY

BY

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Dedication

I dedicate this project first to Almighty Allah, for His blessing and grace over its completion and over every part of the process. I also dedicate it to my supervisor, Miss Ilori Deborah; to my Head of Department, Dr. Ayorinde Oduroye, for his calmness and advice; and to Dr. Pappy Jae, for the time and correction he committed to guiding me. Finally, I dedicate this project to my father, Lamidi Taoheed Ayankojo, and my mother, Lamidi Wakilat Idowu; to my sisters, Taoheed Naheemat and Taoheed Zainab; and to my brothers, Taoheed Abdulrasheed, Lasisi David and Ayowole Abraham.

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Abstract

Nigeria's official economic data is published by several institutions, the Central Bank of Nigeria (CBN), the National Bureau of Statistics (NBS), and the World Bank, in fragmented formats (PDF bulletins, spreadsheets and disparate web portals) that are difficult for students, researchers, developers and the public to access, compare and consume programmatically. This project designed and implemented **NPEDATA**, a web-based platform that aggregates public Nigerian economic data into a single standardised database, presents it through an interactive analytical dashboard, and exposes it through a **free, open REST API** requiring no authentication. The system was built as a seven-stage pipeline (collection, validation, standardisation, storage, analytics, API and presentation) using an iterative prototyping methodology. The backend was implemented in Python with the FastAPI framework and a Supabase (PostgreSQL) database; the frontend is a static dashboard built with HTML, CSS and vanilla JavaScript using Chart.js for visualisation. The API implements hypermedia controls at Richardson Maturity Model Level 3 (HATEOAS). The resulting platform holds **122 indicators** and approximately **12, 100 observations** spanning 1960–2026, including exchange rates, inflation, GDP, monetary policy rate, foreign reserves, multi-currency rates and CBN balance-sheet data. The system was tested through unit tests, functional testing and web accessibility audits. The work demonstrates that fragmented national economic data can be consolidated into an accessible, correct and programmatically consumable resource using free and open-source tools.

Keywords: data aggregation, open data, REST API, HATEOAS, economic analytics, Nigeria, FastAPI, data visualisation.

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CHAPTER ONE, INTRODUCTION

1.1 Background to the Study

Reliable economic data underpins decision-making by government, businesses, researchers and the general public, and its usefulness depends substantially on how accessible and machine-processable it is. Open data, data that anyone may access, use, modify and share, is widely regarded as a public good that converts the cost of publication into broad public value, because each downstream user is spared the effort of repeating the same collection and cleaning work (Open Knowledge Foundation, n.d.; Open Government Working Group, 2007). In Nigeria, the principal producers of official economic statistics are the Central Bank of Nigeria (CBN) and the National Bureau of Statistics (NBS), supplemented by international bodies such as the World Bank. These figures are, however, published across several institutional websites in formats designed for human reading rather than computation, Portable Document Format (PDF) bulletins, individual spreadsheets and web tables whose structure differs from one indicator to the next.

This fragmentation is not merely incidental. Analyses of Nigeria's public-sector information systems attribute the persistence of siloed, non-interoperable data to institutional rather than technical factors, observing that agencies frequently treat the data they hold as a source of value and are consequently reluctant to share it (Eleanya, 2026). Although the Freedom of Information Act has, since 2011, mandated the proactive publication of public information, compliance across federal ministries, departments and agencies remains low (Ogunyale & Osho, 2023). The result is a considerable cost to any user who wishes to combine indicators, since each dataset must be located separately, its date formats and units reconciled by hand, and the series aligned manually before analysis can begin.

The difficulty is compounded by the absence of a programmatic interface. Neither the CBN nor the NBS exposes a public Application Programming Interface (API) for the indicators considered in this study, so that developers and researchers are unable to retrieve the data automatically and must instead re-extract it from published documents. The principle of tidy data, one observation per row, with descriptive metadata held separately, provides a well-established basis for consolidating series of differing frequency and unit into a single analysable structure (Wickham, 2014), yet no freely accessible platform applies it to Nigerian public economic data. It is against this background that the present study designs and develops a web-based platform which aggregates Nigeria's public economic data into a single standardised repository and republishes it through an interactive analytical dashboard for human users and a free, open API for programmatic consumers.

1.2 Statement of the Problem

Public Nigerian economic data is fragmented across several institutional websites and documents, with no common point of access. A substantial proportion of it is published in formats that are not machine-readable, such as PDF bulletins and inconsistently structured spreadsheets, so that every reuse begins with manual extraction. The data is further characterised by inconsistent structure and units: individual series are reported at daily, monthly, quarterly or annual frequencies and are expressed in differing units, including naira thousands, naira millions, percentages and United States dollars, with no unified schema relating them to one another. There is, in addition, no free, documented and authentication-free API through which the data may be queried programmatically, and the existing presentation of the data seldom extends beyond isolated static figures, offering little of the comparison or plain-language interpretation required by non-specialist users. Taken together, these shortcomings place the effective use of Nigeria's public economic data beyond the convenient reach of students, researchers, journalists and independent developers, and give rise to duplicated effort as each user repeats the same collection and cleaning tasks.

1.3 Aim and Objectives

The aim of this study is to design and develop a web-based platform that aggregates Nigeria's public economic data into a single, standardised data store and makes it accessible through an interactive analytical dashboard and a free, open Application Programming Interface (API).

In pursuit of this aim, the study has six specific objectives. The first is to collect public economic indicators from the Central Bank of Nigeria, the National Bureau of Statistics and the World Bank, and to ingest them into a single repository. The second is to design a unified relational data model that standardises indicators, sources and observations irrespective of their frequency or unit. The third is to implement server-side analytical functions, including period change, year-on-year comparison, trend estimation and correlation, over the stored data. The fourth is to develop a free, documented REST API incorporating hypermedia controls (HATEOAS) and requiring no authentication. The fifth is to develop an interactive web dashboard that presents the indicators through clear and accurate visualisations. The sixth is to test and evaluate the platform for correctness, usability and accessibility.

1.4 Scope and Limitations of the Study

The scope of this study encompasses the collection, standardisation, storage, analysis, exposure through an API and visualisation of a defined set of Nigerian public economic indicators. At present the platform holds 122 indicators and approximately 12, 100 observations across the domains summarised in Table 1.1. The dataset constitutes a controlled case study rather than an exhaustive representation of all data published in Nigeria; the architecture is nevertheless designed to accommodate additional data through comma-separated-value (CSV) or API ingestion as it becomes available.

Table 1.1, Data coverage summary

Domain	Frequency	Range	Source
Exchange rate (NGN/USD)	Monthly	2020–2026	CBN
Multi-currency (11 currencies, buy/central/sell)	Monthly	2020–2026	CBN
NFEM daily interbank rates	Daily	2024–2026 (348 sessions)	CBN
Monetary Policy Rate	Per MPC meeting	2020–2025	CBN
Foreign reserves (gross/liquid/blocked)	Monthly	2020–2026	CBN
CBN balance sheet	Monthly	2005–2023	CBN
Annual financial statement	Annual	1960–2012	CBN
Currency in circulation	Monthly	2002–2024	CBN
Inflation (headline/food/core)	Monthly	2003–2026	NBS
GDP growth	Quarterly	2020–2024	NBS
Real GDP by sector (47 sectors)	Quarterly/Annual	1981–2024	NBS
Nominal GDP (USD)	Annual	2020–2024	World Bank

The study is subject to a number of limitations. Data collection is performed manually, by downloading and ingesting published source files, rather than through an automated real-time feed; consequently, the figures presented reflect the most recently ingested snapshot rather than live values. Coverage is bounded by what the source institutions publish, the CBN annual financial statement series, for example, ends in 2012, and cannot be extended by the platform itself. The analytical component employs classical statistical methods, namely correlation, ordinary-least-squares trend estimation and linear forecasting, and does not apply machine-learning techniques; the rationale for this decision is discussed in Section 2.3. Finally, the platform does not attempt automated data collection, does not employ artificial intelligence, and is not presented as official government or national infrastructure; it is a reference implementation developed within the constraints of an academic project.

1.5 Significance of the Study

The study is of value to several categories of user. For students and researchers, it provides a single, comparable and downloadable dataset in place of numerous disconnected sources, thereby reducing the time and effort required before analysis can begin. For software developers, it offers a free and openly documented API upon which third-party applications may be built without authentication or cost. For journalists and members of the public, it presents the data through accessible visualisations accompanied by plain-language interpretation. More broadly, by demonstrating that a standardised and reproducible platform for Nigerian public economic data can be constructed entirely from free and open-source tools, the study provides a practical reference for the wider adoption of open-data practices among the institutions that produce such data.

1.6 Definition of Terms

The principal technical terms used throughout this report are defined in Table 1.2.

Table 1.2, Definition of terms

Term	Definition
Aggregation	The collection of data from multiple sources into a single store.
API (Application Programming Interface)	A defined means by which software programs request and exchange data.
REST	An architectural style for web APIs based on standard HTTP methods and resources.
HATEOAS	Hypermedia as the Engine of Application State; the embedding, within REST responses, of links that direct a client to related resources.
CBN	Central Bank of Nigeria.
NBS	National Bureau of Statistics.
NFEM	Nigerian Foreign Exchange Market.
MPR	Monetary Policy Rate; the benchmark interest rate set by the CBN.
Indicator	A measured economic series, for example inflation.
Observation	A single value of an indicator on a specified date.

CHAPTER TWO, LITERATURE REVIEW

2.1 Introduction

Before the design presented in Chapter Three could be settled, it was necessary to confirm that the problem is real rather than a matter of personal inconvenience, and that the proposed solution does not merely reinvent something already in existence. This chapter provides that groundwork. It works through the ideas the platform rests on, open data, aggregation and standardisation, data quality, REST and hypermedia, honest visualisation, and the classical statistics behind the analytics layer, then states the theoretical framework I adopted, reviews the systems already publishing economic data (both international and Nigerian) to see where they fall short, and closes with the technology choices the implementation is built from. By the end of it, the gaps that Chapter Three's design has to answer should be concrete rather than assumed.

2.2 Conceptual Review

2.2.1 Open data and open government data. Open data is data anyone can access, use, modify and share for any purpose, at most subject to attribution (Open Knowledge Foundation, n.d.). For government data specifically, the widely cited principles hold that it should be complete, primary, timely, accessible, machine-processable, non-discriminatory, non-proprietary and licence-free (Open Government Working Group, 2007). Two of those do most of the work here. Machine-processability matters because a PDF table is technically "published" while being practically closed to computation. Accessibility matters because data spread across fragmented portals carries a real cost to reach even when it is nominally free. The economic case for open data, in short, is that it turns publication from a cost centre into public value, every downstream user, researcher, journalist or start-up, stops repeating the same cleaning work someone before them already did. That is essentially the same argument the Nigerian scenario in Section 3.3 makes, just stated more formally.

2.2.2 Data aggregation, standardisation and tidy data. Gathering series from disparate sources is only useful once they are standardised into a common structure, and for that the study adopts the "tidy data" principle (Wickham, 2014): one observation per row, one variable per column, with metadata such as unit, source and frequency kept separate from the values themselves. What this buys in practice is that series of any frequency at all, daily NFEM fixings, monthly CPI, quarterly GDP, even the CBN's 1960–2012 annual financial statement, can sit in one relational table and be queried by the same engine. The alternative, a bespoke table per dataset mirroring each source file's own layout, would just reproduce the fragmentation this project is trying to remove, only now inside the database instead of across it. The relational model itself (Codd, 1970) is what makes that safe: typed columns, foreign keys tying observations back to indicator metadata, and a uniqueness constraint that makes duplicate ingestion structurally impossible rather than merely discouraged.

2.2.3 Data quality. The data-quality literature usually judges datasets against accuracy, completeness, consistency, timeliness and validity, and three of those mattered a great deal here. On consistency: the same institution often publishes related series in different units, the CBN's monthly balance sheet in thousands of naira, its annual statement in millions, so a platform merging them has to carry that unit metadata explicitly, or it will silently mislead. Chapter Four documents a case where exactly this went wrong before it was caught. On accuracy and validity: aggregation multiplies the places an error can creep in, which is why validation became a first-class pipeline stage rather than an afterthought, and why it is unusually exposed as a public service anyone can try. On completeness: no source is complete, services-sector GDP, for instance, simply ends earlier than its sibling series, and the response adopted here is to state that coverage plainly rather than obscure it.

2.2.4 Web APIs, REST and the Richardson Maturity Model. Representational State Transfer (REST), introduced by Fielding (2000), remains the dominant architectural style for web APIs: data as resources addressed by URLs, manipulated with standard HTTP verbs. The Richardson Maturity Model (Richardson & Ruby, 2007; Fowler, 2010) grades how far an API actually goes with that idea, across three levels. Level 1 introduces distinct resources; Level 2 uses HTTP verbs and status codes properly; Level 3, HATEOAS (Hypermedia As The Engine Of Application State), embeds links in every response so a client can discover related actions instead of hard-coding URLs. Very few APIs bother reaching Level 3, but it is disproportionately useful for an open API whose consumers are strangers, because the API ends up describing itself, navigable from the root with no documentation required. For non-JSON responses the equivalent mechanism is the standard Link header (Nottingham, 2017). I implemented Level 3 end-to-end here, and unusually, built an interactive way to see it work (the HATEOAS Explorer in Chapter Four) rather than just asserting it in a spec.

2.2.5 Honest data visualisation. At its core, the visualisation literature is a literature about not misleading people. Tufte (1983) argued for maximising the share of ink that actually carries data and against decoration that distorts it, and a long line of practitioner criticism singles out dual-axis charts as a particular hazard, since two independent y-scales allow a designer to manufacture or exaggerate a correlation simply by sliding one scale. Anscombe (1973) made the underlying point unavoidable with four datasets that share identical summary statistics yet look entirely different when plotted, summary numbers require a trustworthy plot alongside them, not in place of one. These findings are translated into concrete design rules rather than general intentions: no dual axes appear anywhere on the platform (differing scales are shown instead as aligned panels or standardised z-scores), axes are never truncated to exaggerate a movement, units are always stated, and every chart carries a plain-language note, since a technically correct chart that a non-specialist cannot interpret is only partially honest.

2.2.6 Statistical foundations of the analytics layer. The analytics on this platform are deliberately classical rather than fashionable. The Pearson product-moment correlation coefficient measures linear association between paired series, and its significance is assessed with a two-tailed Student-t test computed via the regularised incomplete beta function (Press et al., 2007). Ordinary least squares supplies the trend line, reported alongside R^2. Two cautions that the statistics literature usually leaves as footnotes are instead built into the product itself: correlation is not causation, and two series that both simply trend over time will correlate even when nothing connects them (Granger & Newbold, 1974), so the platform recomputes correlation on first differences and warns the user when that detrended figure collapses relative to the headline one. Significance is also kept visually separate from strength, since the two are easily conflated: a weak correlation can still be highly significant in a large enough sample, and the interface is designed to preserve that distinction.

2.3 Theoretical Framework

I organised both the design and the evaluation of this project around two established frameworks rather than inventing my own criteria as I went.

The first is the FAIR data principles: data should be Findable, Accessible, Interoperable and Reusable (Wilkinson et al., 2016). FAIR is the yardstick for the data side of the platform specifically. Findability is served by one catalogue of 122 indicators with searchable metadata; accessibility by a free dashboard and a no-authentication API; interoperability by one tidy schema with ISO-8601 dates and stated units throughout; reusability by CSV export, citation generation, provenance fields and a reproducible seed snapshot.

The second is the Richardson Maturity Model from Section 2.2.4, which does the equivalent job for the interface side: the Open API was designed to reach Level 3, and Chapter Four verifies that it actually does rather than just claiming it.

Between the two, the underlying thesis of the project is straightforward: fragmentation is not solved by building yet another website, but by making the data itself FAIR and making its interface hypermedia-driven.

2.4 Review of Related Systems

FRED (Federal Reserve Economic Data, St. Louis Fed). This is the reference point for what a national economic-data platform can look like: hundreds of thousands of series, a documented API, charting, downloads, citations, all of it. Its relevance to Nigeria is thin though, since Nigerian coverage is limited to coarse international aggregates. What FRED really demonstrates is the category this project belongs to, while underlining that no Nigerian equivalent of it exists yet.

World Bank Open Data. Fully open, with a long-standing public API and standardised indicator codes, and this project both reviews it and consumes it directly for nominal GDP. Its limitation is granularity, mostly annual, country-level aggregates, published with a lag, so it simply cannot answer intra-year questions like monthly inflation dynamics or daily FX behaviour, which is exactly where the domestic sources have to take over.

IMF Data. Authoritative macroeconomic and financial statistics with programmatic access, but like the World Bank it's oriented toward cross-country aggregates rather than the granular domestic series (NFEM daily fixings, individual CBN balance-sheet lines) this project carries.

Our World in Data. Not an API platform, but probably the strongest model available of explanatory data publication: every chart sits inside plain-language narrative, with sources and methods disclosed alongside it. I borrowed its storytelling pattern, what happened, why it matters, how to read it, for every indicator page on this platform. Nigerian macroeconomic coverage there is, again, limited to international aggregates.

Trading Economics and Statista. Broad commercial aggregators with polished interfaces, both largely paywalled, and both ultimately re-selling the same CBN and NBS publications underneath. If anything they're proof that commercial demand exists for exactly the kind of aggregation this project performs, it's just that their pricing model becomes part of the access problem for the students and journalists this project is actually aimed at.

Central Bank of Nigeria (CBN) publications. The authoritative source for monetary and financial data, and the clearest illustration of the machine-readability gap this project responds to: statistics spread across web pages, PDF bulletins and per-topic Excel files, with layouts and units that vary between documents and no public API in sight (Section 3.3 walks through the workflow this forces on anyone trying to use it).

National Bureau of Statistics (NBS). The authoritative source for CPI and GDP. Publications are report-oriented, PDF with accompanying tables, series get rebased periodically, and there's no unified programmatic interface for any of the indicators this project covers.

data.gov.ng and Nigerian open-data initiatives. Nigeria does have an official open-data portal and has taken part in international open-government initiatives, and civic-tech organisations, BudgIT in particular, which visualises public budgets, show there's a real domestic appetite for usable public data. But these efforts centre on budgets, spending and static dataset publication rather than continuously usable, API-accessible economic time series, which is the specific niche this project sits in.

Why has the Federal Government not already built this? This is a fair question, and the answer is, on balance, not a technical one. Nigeria has had an official e-Government Interoperability Framework since 2018, setting out exactly how ministries, departments and agencies are supposed to exchange data with each other (National Information Technology Development Agency [NITDA], 2018), and adoption has still lagged badly enough that, as of 2026, government databases as fundamental as the National Identity Management Commission's NIN registry and the CBN's own BVN system still don't talk to each other (Eleanya, 2026). The reasons people who actually work on this problem give are institutional, not technological: agencies tend to see the data they hold as a source of value and influence, and sharing it can feel like giving that up, while different agencies also classify and manage the same kind of information differently, which makes harmonising it harder than it needs to be (Eleanya, 2026). BudgIT's own analysis is blunter still, arguing that Nigeria's fragmented systems persist "not because of an engineering failure" but because fragmentation is useful to whoever benefits from the resulting opacity, and that fixing it is a governance choice before it is a technical one (Anintah, n.d.). A final-year project cannot resolve that governance problem, and no such claim is made here. What a project of this kind can do is remove the technical excuse: it demonstrates that one student, with no budget and no institutional authority, can stand up a working, standardised, publicly queryable version of this data using entirely free tools within an academic year. If that is achievable at this scale, the claim that unifying Nigeria's economic data is infeasible does not hold, what is absent is the will to do it, not the means.

Is this, then, a legislative problem or an institutional one? It is both, and the answer does not simplify neatly. There is a real legislative gap: Nigeria's Freedom of Information Act has legally mandated proactive publication of exactly this kind of public data since 2011, yet a six-organisation civil-society audit, including BudgIT and the International Centre for Investigative Reporting, found that of 250 federal MDAs surveyed, 153 still scored below a basic compliance benchmark in 2022 alone (Ogunyale & Osho, 2023). That is not a case of no rule existing; it is a case of a rule existing and being routinely ignored, which points to enforcement rather than legislation as the deeper gap. Nigeria has closed an analogous gap before, though. The 2019 data-protection framework (the NDPR) relied on NITDA, a technology-development agency with no actual legal power to penalise anyone, to police compliance, and when the Nigeria Data Protection Act replaced it in 2023, it created an independent regulator, the Nigeria Data Protection Commission, with real investigative and fining powers (Falore & Jidda, 2026). No equivalent body exists yet for government-data interoperability specifically: Ne-GIF is guidance, not law, and ignoring it carries no consequence, the same way ignoring the FOI Act mostly hasn't. If there is a legislative fix available here, on this evidence it probably isn't another framework document, Nigeria already has one of those, but a body with the same enforcement teeth the NDPC was given, empowered specifically to make interoperability mandatory rather than merely recommended.

2.5 Gap Analysis

Table 2.1, Feature-level gap analysis

Capability	**FRED**	**World Bank**	**Trading Econ.**	**CBN**	**NBS**	**NPEDATA**
Granular Nigerian series (daily/monthly)	No	No	Partial	Source	Source	**Yes (aggregated)**
Free, no-authentication API	Key req.	Yes	No	No	No	**Yes**
Hypermedia	No	No	No	,	,	**Yes**

(HATEOAS L3) API						
One standardised schema across sources	Yes	Yes	Yes	No	No	**Yes**
Correlation with significance (R^2, p)	No	No	No	,	,	**Yes**
Spurious-correlation warnings	No	No	No	,	,	**Yes**
Plain-language interpretation per series	No	Partial	No	No	No	**Yes**
Public validation-as-a-service	No	No	No	,	,	**Yes**
Citable exports (CSV/PNG/APA)	Yes	Yes	Partial	No	No	**Yes**

A clear pattern emerges once the comparison is laid out in a table: the systems with excellent access lack granular Nigerian data, and the systems that hold the Nigerian data lack machine access. More specifically, none of the systems reviewed, at any scale, combine statistical-integrity features, significance testing and spurious-correlation detection, with plain-language interpretation for a non-specialist reader. It is that combination, not any single feature, that this project fills.

2.6 Review of Enabling Technologies

Choosing the stack meant weighing several genuine alternatives rather than defaulting to whatever was familiar; the choice made at each layer, and what was set aside, is set out below.

For the **API framework**, FastAPI was chosen over Flask and Django REST Framework. It gives automatic OpenAPI/Swagger documentation and Pydantic request validation almost for free, which matters more than it might sound, since the ingestion layer's type checks ride on the same validation, plus async support, all at a fraction of Django's footprint. An auto-documented API isn't just a nice-to-have here; the API is itself one of the two deliverables, so documenting it well is part of the product, not an add-on.

For the **database**, PostgreSQL via Supabase, over MySQL, MongoDB and Firebase. The data is inherently relational, sources feed indicators feed observations, and integrity constraints are doing real work in that chain, which argues against a document store from the start. Supabase adds a managed free tier and an auto-generated REST layer with row-level security on top, which allows the dashboard to read the database directly and safely without a hand-rolled API in between, while remaining standard PostgreSQL underneath should migration off it ever be required.

For **visualisation**, Chart.js v4 over D3.js and the commercial options. D3 gives unlimited control but at a steep cost in code volume for what are, mostly, standard chart types here; the commercial libraries conflict with the project's open ethos on principle. Chart.js's plugin system turned out to be enough for the honesty-driven customisations the project actually needed, annotations, crosshairs, end-of-line labels.

For the **frontend**, static HTML, CSS and vanilla JavaScript over React or Vue. No build step, free static hosting, and full view-source auditability, anyone can open the page source and see exactly what's running, which fits an "open" project better than a bundled framework output would. The known weakness of this approach, code drifting apart across pages, is given a deliberate mitigation (a single shared library) that is assessed quantitatively later, in Section 4.9.

For **hosting**, GitHub Pages, Render and Supabase together, which is an entirely free and reproducible deployment. It does have a real trade-off, the free-tier API has cold starts, and rather than hide that, Section 4.9 measures it and shows what was done about it.

Finally, for **machine-consumer interfaces**, support was added for MCP and llms.txt. The Model Context Protocol allows AI assistants to call the API as tools, and llms.txt provides machine-readable platform discovery, extending "open" a step further than human developers alone, to automated consumers as well.

2.7 Summary

Four conclusions carry forward into the design in Chapter Three. First, open, machine-processable economic data is under-supplied in Nigeria, because the authoritative sources publish for reading rather than for computation. Second, the conceptual tools needed to address this already exist and are mature, tidy data for standardisation, FAIR for openness, REST Level 3 for the interface, so the task is one of disciplined application rather than invention. Third, the visualisation and statistics literature provides concrete integrity rules (no dual axes, significance kept separate from strength, a detrending check on correlations) that can be built directly into a product rather than left as caveats in a footnote. Fourth, none of the systems reviewed, Nigerian or international, combines granular Nigerian coverage, open machine access and built-in statistical integrity in one place. Chapter Three sets out to design the system that does.

CHAPTER THREE, SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter translates the problem stated in Chapter One and the review in Chapter Two into a design. It sets out the methodology adopted and the reasons for selecting it over the alternatives, describes how Nigerian public economic data is obtained at present through a concrete scenario rather than an abstract description, specifies the proposed system and its requirements, and then develops the design in detail, architecture, use cases, data flow, database design and data dictionary, API design, core algorithms, input/output design, security, and the user interface. Care is taken throughout to relate each decision to a specific reality of the Nigerian data landscape rather than presenting it as an abstract choice, since very few of the decisions were abstract.

3.2 Research and Development Methodology

The system was built iteratively and incrementally rather than planned in full at the outset. The data model and ingestion scripts came first, followed by the Open API, the dashboard pages, the analytics engine, and finally a set of refinement passes on visualisation integrity, accessibility and the different stakeholder views. Each increment was reviewed and checked against the stored data before the next was begun.

Why not another approach. Two other candidate methodologies were considered before this one was adopted.

Table 3.0, Methodology comparison

Methodology	Strength	Why it was unsuitable here
Waterfall	Strong documentation discipline; predictable phases	Assumes requirements are fully known up front, and mine weren't, the structure, units and quirks of CBN/NBS publications only became clear once I was actually ingesting them (the CBN balance sheet, for instance, is published

		in ₦'000 while its annual statement uses ₦ millions). A frozen upfront specification would simply have been wrong by the second week.
Scrum/agile (team-oriented)	Responsive to change; strong feedback cadence	Built around a multi-person team with defined roles, product owner, scrum master, and so on. Running that ceremony solo would have been overhead for its own sake.
Iterative prototyping (chosen)	Working software early; each cycle absorbs what the previous one taught	Fit a solo developer, data sources that kept revealing new quirks, and a supervisor-feedback loop naturally. Every iteration ended with what I came to think of as a data-truthfulness check, going back and re-verifying that what the screens claimed actually matched what the database held.

The habit maintained throughout the process was treating verification as part of every iteration rather than reserving it for the end: after each increment, the figures displayed by the frontend were cross-checked against the database itself, and any discrepancies (several of which Chapter Four documents) were corrected before new work began. This is why this chapter and the testing chapter are so closely connected.

3.3 Analysis of the Existing System

The term "existing system" here refers not to a single piece of software, none exists, but to the manual workflow that any user of Nigerian public economic data must follow today, shaped by how three institutions each publish their figures:

The Central Bank of Nigeria (CBN) publishes exchange rates, money and credit statistics and its Statistical Bulletin through its website, mostly as PDF documents and per-topic Excel downloads; different datasets sit on different pages, with different layouts, date formats and units that change between publications, thousands of naira in one table, millions in the next, and there is no public API. The National Bureau of Statistics (NBS) publishes the monthly Consumer Price Index and quarterly GDP reports as PDF documents with spreadsheet tables attached; the sector-GDP tables alone run to dozens of columns, are rebased periodically, and are laid out for reading rather than for computation, and again expose no public API. The World Bank does offer a proper API for its indicators, but its Nigerian coverage is coarse, mostly annual, national-level aggregates, so it cannot substitute for the domestic sources where monthly or daily granularity is required.

A concrete scenario. Consider a final-year economics student in Lagos attempting to answer a question that is actively debated: how did the June 2023 foreign-exchange unification relate to the inflation surge that followed it? Under the existing system, the student must locate the CBN exchange-rate page and download the monthly rates, separately locate the NBS CPI report archive and extract headline inflation month by month, reconcile the two by hand across differing date formats and file layouts, align them in a spreadsheet and compute a correlation manually with no guidance on whether the result is statistically meaningful, and then repeat the entire exercise each time a new month is published. In practice this represents hours, sometimes days, of preparation before analysis can begin; every manual step is an opportunity for error; and the exercise is effectively out of reach for a journalist working to a deadline, a secondary-school teacher, or a developer who requires the figures inside a program.

Table 3.0b, Weaknesses of the existing workflow

#	Weakness	Consequence
1	Data scattered across institutions and pages	High search cost; no single point of access
2	PDF/spreadsheet formats designed for reading	Not machine-readable; manual re-typing and extraction
3	Inconsistent date formats, layouts and units	Reconciliation errors; silent unit mistakes (₦'000 vs ₦m)
4	No public API at CBN/NBS	Programmatic and third-party use effectively impossible
5	No built-in analysis or interpretation	Non-experts cannot judge trends or correlation reliability
6	Work is repeated by every user, every month	Nationally duplicated effort; no shared cleaned dataset

3.4 Analysis of the Proposed System

NPEDATA replaces that manual workflow with a maintained pipeline behind two access paths, an interactive dashboard for people, and a free Open API for programs.

The same scenario, replayed on NPEDATA. The same student now opens the Compare Indicators page, picks Headline Inflation and Exchange Rate NGN/USD, and clicks Compare. In under a minute she has both series date-aligned automatically, a single honest chart (z-score standardised rather than a misleading dual axis), the Pearson correlation together with its R^2 and statistical-significance p-value, a warning if the relationship looks like a shared trend rather than a real association, the full paired data table, and a citable CSV export. Should a document be required, Briefing Studio composes a cited, print-ready brief of the same indicators. Because this particular question, how the reform relates to what followed, is one that is actively debated rather than merely studied, the dedicated Reform Impact page addresses it directly: every headline indicator is split into a before-June-2023 average and an after-June-2023 average, computed live from the same database, with neither side of the argument favoured in the presentation. A critic of the reform and a supporter of it can both point to real figures on that single page (Section 4.6, Figure 4.15). What previously required days of error-prone preparation now takes minutes; the statistical caveats are supplied rather than left to chance; and the platform itself takes no side in the argument for which its figures are used.

Table 3.0c, Existing vs proposed system

Dimension	Existing workflow	NPEDATA
Access	Many sites, many files	One dashboard + one API
Format	PDF/Excel for reading	Standardised machine-readable series
Units/dates	Inconsistent, error-prone	One schema: ISO dates, stated units
Programmatic use	None (CBN/NBS)	Free REST API, HATEOAS Level 3, no key
Analysis	Do-it-yourself	Built-in stats with significance + honesty guards
Interpretation	None	Plain-language storytelling per

		indicator
Cost & repetition	Every user repeats the work	Cleaned once, shared by all

3.5 System Requirements

Requirements were gathered from the scenario analysis in Section 3.3, the project objectives in Chapter One, and supervisor feedback across iterations. They are stated per stakeholder group where relevant.

Table 3.1, Functional requirements

ID	Requirement	Primary stakeholder
FR1	Ingest indicators and observations from CSV/Excel source files into the database	Maintainer
FR2	Validate incoming data (ISO dates, numeric finite values, duplicates, future dates) before storage	Maintainer / data quality
FR3	Standardise all series, any frequency, any unit, into one observations schema	All
FR4	Serve every indicator's series through documented REST endpoints, no authentication	Developers
FR5	Embed hypermedia controls (_links, RFC 8288 Link header) so the API is navigable from the root (HATEOAS Level 3)	Developers
FR6	Provide per-indicator analytics for the full catalogue: latest, period change, year-on-year, range, mean, volatility, OLS trend with R^2 and p-value, illustrative forecast	Researchers
FR7	Provide cross-indicator comparison with Pearson r, R^2, significance p-value, and reliability warnings (short overlap, mixed frequency, shared-trend/detrended check)	Researchers
FR8	Display interactive charts with plain-language storytelling (what happened / why it matters / how to read it)	Public / students
FR9	Allow date-range filtering, range presets (1Y/3Y/5Y/All), and event-context markers on charts	All
FR10	Export any indicator as CSV; download any chart as an attributed image; generate an APA citation	Researchers / students
FR11	Offer two reading depths, plain-language Reader view and statistics-rich Analyst view, from one codebase	All
FR12	Compose a cited, print-ready multi-indicator briefing, shareable as a regenerating link	Journalists / policymakers
FR13	Expose the validation layer as a public service returning per-row verdicts,	Developers / demonstration

	guaranteed never to write	
FR14	Provide embeddable live chart widgets for third-party sites	Publishers
FR15	Provide machine-readable platform discovery for AI agents (llms.txt; MCP server)	AI agents

Table 3.1b, Non-functional requirements (with measurable targets)

ID	Requirement	Target	Achieved (evidence in Ch. 4)
NFR1	Correctness	Every displayed figure matches the stored data exactly	Systematic audit; defects found were fixed and documented
NFR2	Accessibility	WCAG 2.1 AA contrast (≥ 4.5:1)	Lighthouse accessibility 100/100
NFR3	Performance	Interactive charts on consumer hardware; API responses in low seconds when warm	Warm endpoints ≈0.7–3 s; repeat reads cached to ≈1 s
NFR4	Availability	Publicly hosted, free to access	GitHub Pages + Render + Supabase, all free tiers
NFR5	Honesty	No misleading visual encodings; uncertainty stated	Single-axis policy, significance reporting, warning system
NFR6	Portability/reproducibility	Rebuildable from the repository alone	Seed snapshot + setup.sql + pinned dependencies
NFR7	Robustness	Malformed input never corrupts data or crashes the API	Adversarial test suite (incl. NaN/∞, 5, 000-row floods) passes

Hardware and software requirements. Development: a standard laptop, Python 3.10+, a modern browser, internet access. Deployment: GitHub Pages (static frontend), Render (FastAPI service), Supabase (managed PostgreSQL). End users need only a browser, including on mobile; API consumers need any HTTP client.

3.6 System Architecture

The system is organised as a seven-stage pipeline (Figure 3.1).

Figure 3.1, Seven-stage architecture

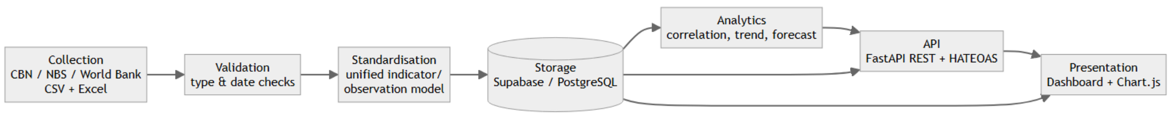


Figure 3.1, Seven-stage system architecture

Each stage performs a distinct function within this project. In the **collect** stage, source files are obtained from the three institutions' published outputs, CBN statistical pages, NBS CPI and GDP report tables, and World Bank indicators, as CSV or Excel. In the **validate** stage, loader scripts type-check every row, normalise dates to ISO 8601, and reject malformed records; during the build this stage caught and removed 1, 435 duplicate records introduced by repeated ETL runs, reducing 13, 535 raw rows to the 12, 100 clean observations in production, which is evidence that the stage performs real work. In the **standardise** stage, every series, whether daily NFEM rates or the 1960–2012 annual financial statement, is reduced to the same (indicator_id, obs_date, value) shape, with unit and source held as metadata. In the **store** stage, the data resides in Supabase-hosted PostgreSQL with a uniqueness constraint preventing re-duplication, and public read access is gated by row-level security. In the **analyse** stage, classical statistics (descriptives, OLS trend, and Pearson correlation with a Student-t significance test) are computed in the API and, for interactive pages, in the browser from the same data. In the **API** stage, a FastAPI service exposes the catalogue as versioned REST with hypermedia controls, its ingestion endpoints demo-safe by default. Finally, in the **present** stage, the static dashboard deliberately reads the database's REST layer directly, so that the primary user experience does not depend on the API host being awake.

Figure 3.1b, Deployment view

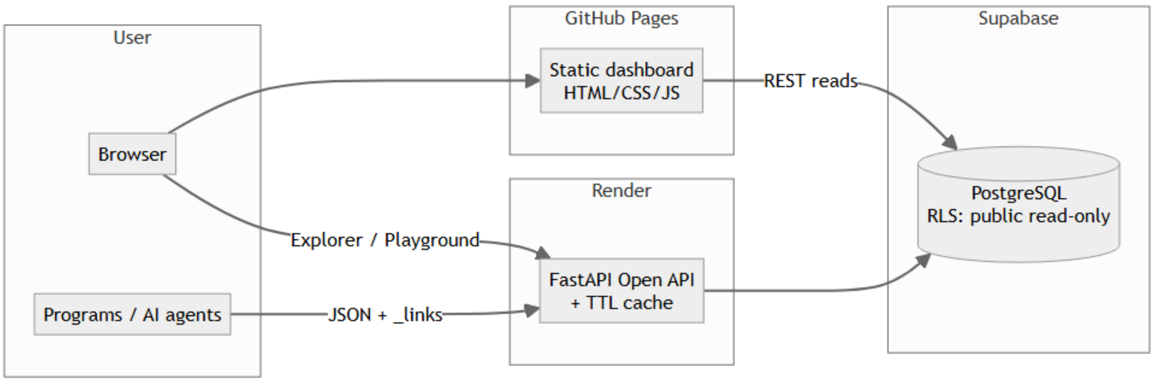


Figure 3.1b, Deployment view

The two data paths are intentional: the dashboard's independence from the API host is an availability decision (assessed further in Section 4.9).

3.7 System Design

3.7.1 Use-case design (Figure 3.2)

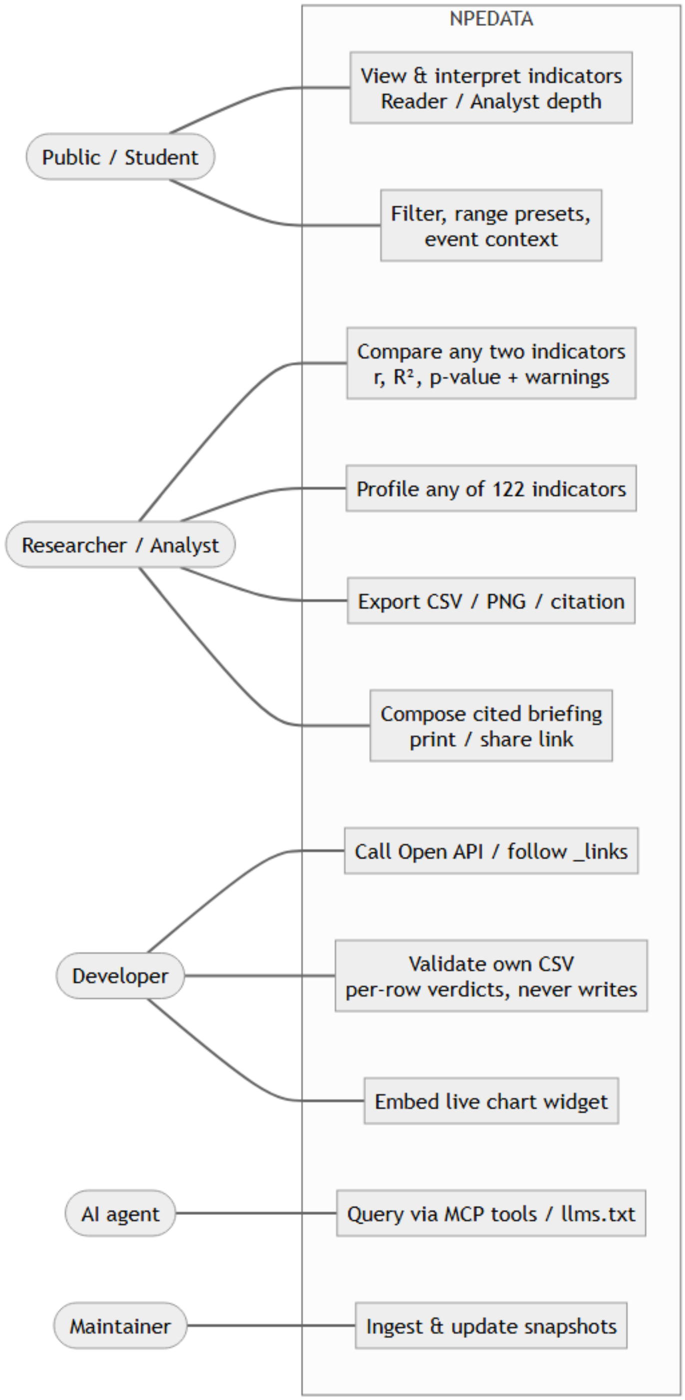


Figure 3.2, Use-case diagram

Table 3.1c, Use-case descriptions (three representative cases in full)

UC-1: Interpret an indicator (Public/Student). *Precondition:* none, public site. *Main flow:* (1) user opens an indicator page, e.g. Inflation; (2) system fetches the series and renders the headline stat, the plain-language story blocks, and the chart with event markers; (3) user narrows the window with a range preset; (4) optionally flips the navbar dial to *Analyst*, revealing the statistical panel (range, mean, σ, OLS trend with R^2 and p-value). *Postcondition:* none (read-only). *Alternative flow:* data unavailable → a labelled error state with retry, never a blank chart.

UC-2: Test a correlation hypothesis (Researcher). *Precondition:* none. *Main flow:* (1) researcher opens Compare Indicators and selects any two of the 122 indicators; (2) system date-aligns the two series on their common observations; (3) system computes Pearson r, R^2, and a two-tailed p-value, and plots both series z-score-standardised on one axis; (4) system runs the reliability guards, if the overlap is short, the frequencies differ, or the detrended (month-to-month change) correlation collapses relative to the level correlation, a visible warning explains the caveat; (5) researcher exports the paired table as CSV or copies the citation. *Postcondition:* none. *Alternative flow:* no overlapping dates → explanatory notice, no correlation shown.

UC-3: Validate a dataset via the API (Developer). *Precondition:* developer has a CSV with obs_date, value columns. *Main flow:* (1) client POSTs the file with a target indicator_id to /api/v1/validate/csv (or uses the Pipeline Playground UI); (2) system checks the header, then judges every row, ISO date, numeric and finite value, no in-file duplicate date, no future date; (3) system returns a per-row verdict report with reasons and normalised values, plus written_to_database: false. *Postcondition:* **no state change ever**, the endpoint is validation-only by design. *Alternative flows:* unknown indicator → 404 listing valid ids; non-UTF-8 file or missing columns → 400 with the specific message.

3.7.2 Data-flow and interaction design. At Level 0, external sources supply raw data to the NPEDATA process, which stores standardised observations and returns charts, JSON and CSV to users. At Level 1 the process decomposes into *Ingest* → *Validate* → *Store* → *Query* → *Analyse* → *Serve*. Two representative interactions are shown as sequence diagrams.

Figure 3.3, Sequence: loading an indicator page

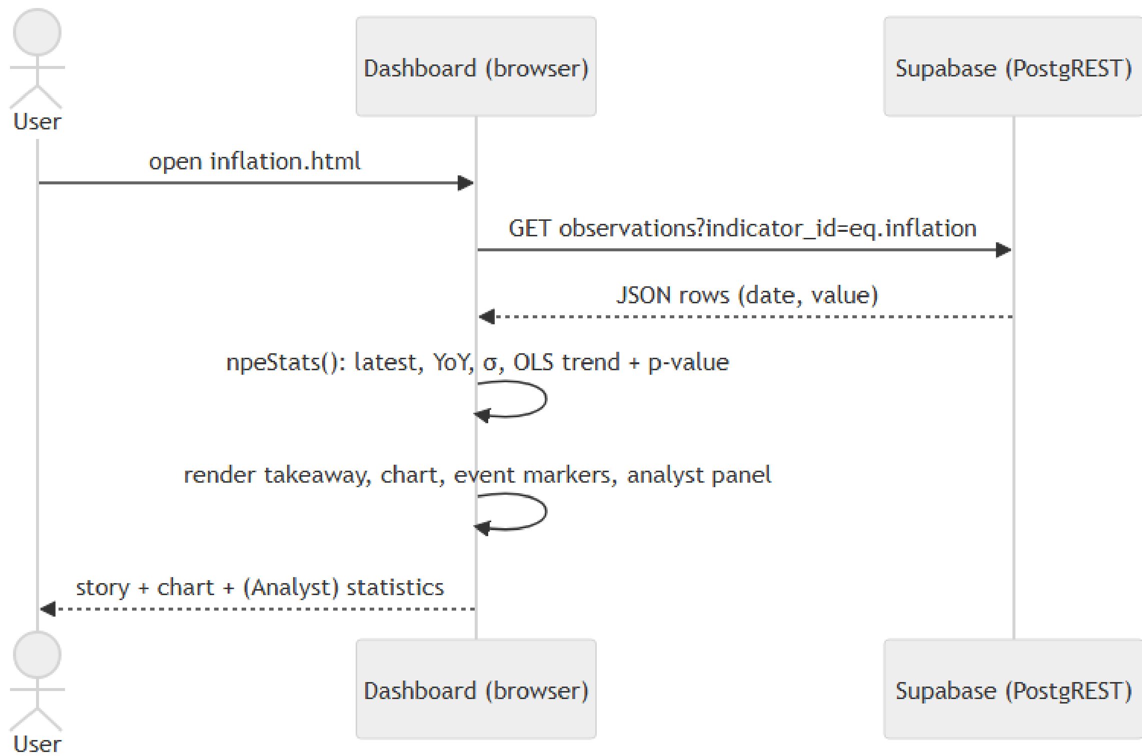


Figure 3.3, Sequence diagram: loading an indicator page

Figure 3.4, Sequence: validating a CSV through the Open API

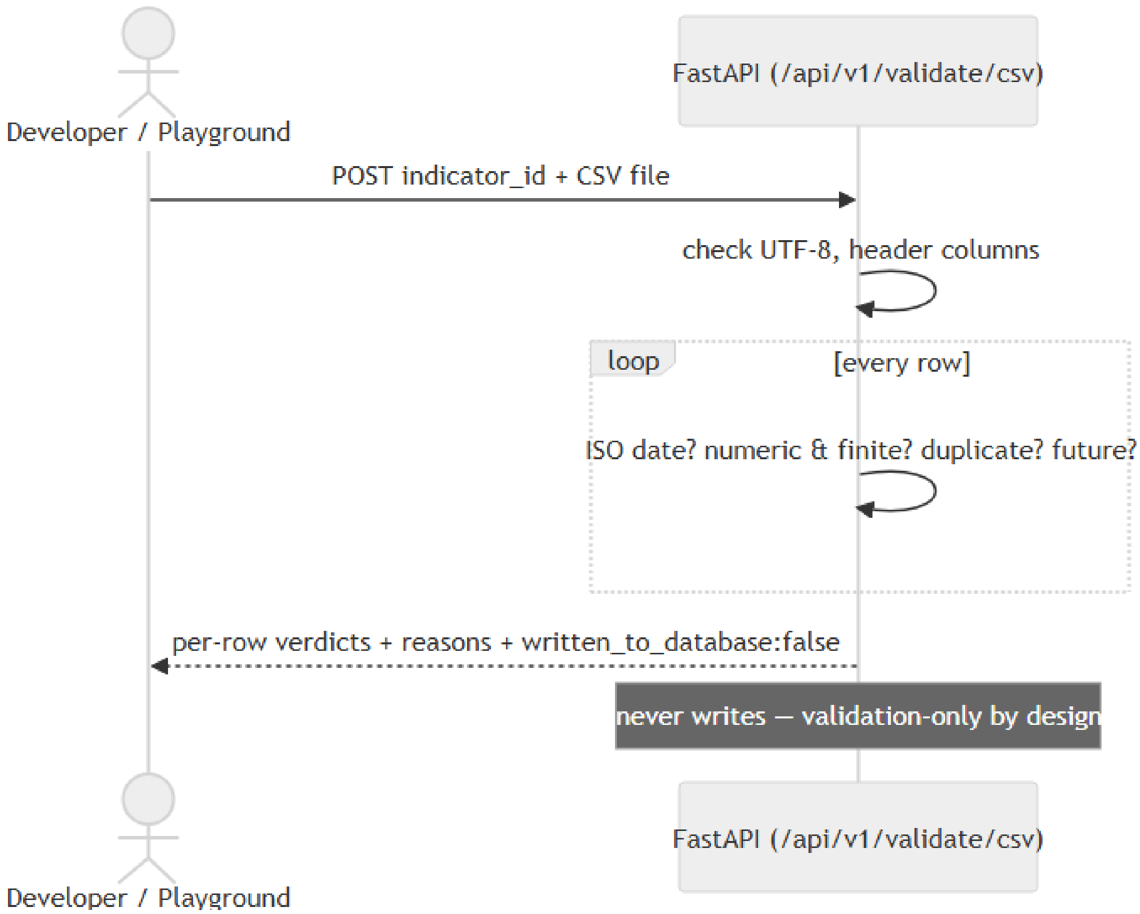


Figure 3.4, Sequence diagram: validating a CSV through the Open API

3.7.3 Database design (Figure 3.5, ERD)

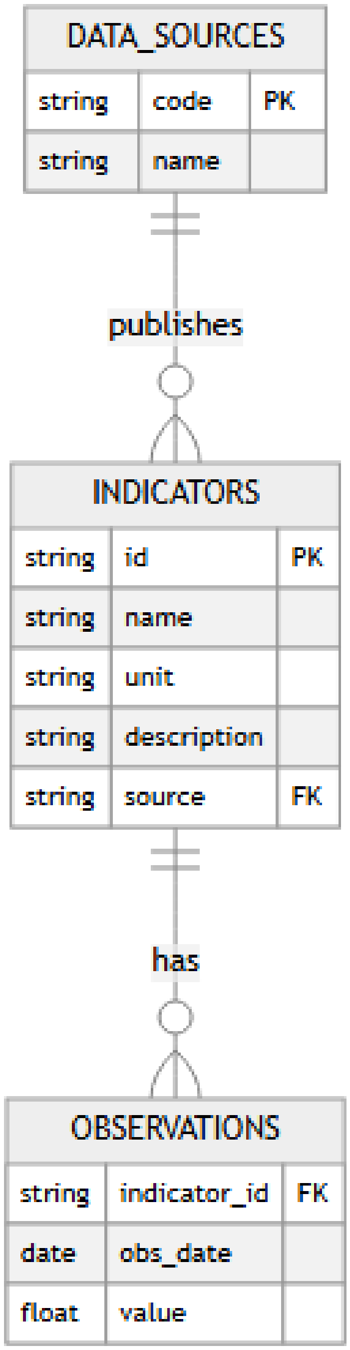


Figure 3.5, Entity-relationship diagram

Table 3.2, Database schema (core tables)

Table	Key columns	Purpose
data_sources	code, name	The publishing institutions (CBN, NBS, World Bank)
indicators	id, name, unit, description, source	Metadata for each series
observations	indicator_id, obs_date, value	One value per indicator per date (tidy/long form)

The long/tidy observations table lets indicators of any frequency or unit coexist in one structure, the key standardisation decision of the project.

Table 3.2b, Data dictionary

Table data_sources

Field	Type	Constraints	Description	Example
code	text	PK	Short institution code	CBN
name	text	not null	Full institution name	Central Bank of Nigeria

Table indicators

Field	Type	Constraints	Description	Example
id	text	PK, snake_case	Stable series identifier	exchange_rate
name	text	not null	Human-readable name	Exchange Rate NGN/USD
unit	text	not null	Unit **as stored** (critical, see note)	Naira per USD
description	text	,	What the series measures	…
source	text	FK → data_sources.code	Publishing institution	CBN

Table observations

Field	Type	Constraints	Description	Example
indicator_id	text	FK → indicators.id; unique with obs_date	Which series	inflation
obs_date	date	ISO 8601; unique with indicator_id	Observation date	2024-12-01
value	numeric	not null, finite	The observation, in the indicator's stored unit	34.80
source	text	,	Provenance tag for the row	NBS

Unit note (a genuine design lesson of this project): stored units differ by series, the CBN monthly balance sheet is in ₦'000, its annual statement in ₦ millions, GDP sectors in ₦ billions. The presentation layer therefore carries a single scale-aware formatter so that values are always displayed accurately (for example, ₦81.04T); Chapter Four documents the defects this discipline caught.

3.7.4 API design. The API is versioned under /api/v1/ and returns JSON with an embedded _links object (HATEOAS). Representative endpoints include /summary, per-indicator series endpoints (/gdp, /inflation, /exchange-rate, /fx-reserves, /nfem, /multicurrency, …), /analytics/{indicator_id}, /coverage, /export/{indicator_id} (CSV, with an RFC 8288 Link header), and /validate/csv (the validation layer as a service). Four design rules govern every endpoint: (1) **versioned paths** so future changes cannot break consumers; (2) **hypermedia everywhere**, every JSON response includes _links (self, index, docs, related resources, per-indicator analytics/export), making the whole API navigable from the root; (3) **no authentication, open CORS**, the mission is access; (4) **writes are demo-safe by default**, ingestion validates and normalises but persists nothing unless the server-side ALLOW_DATA_WRITES flag *and* an explicit commit=true are both set.

3.7.5 Algorithm design. The three core computations, as implemented:

Validation (per CSV row):

```
for each row (r = 2, 3, …):
 problems ← []
 if obs_date does not parse as YYYY-MM-DD: problems += "ISO date required"
 if value does not parse as a number: problems += "numeric required"
 else if value is NaN or ±Infinity: problems += "finite number required"
 if obs_date parsed:
 if obs_date > today: problems += "future date"
 if obs_date already seen in this file: problems += "duplicate date"
 emit {row, status: valid|rejected, reasons, normalized}
```

```
never write; return counts + verdicts
```
Pearson correlation with significance:

```
align the two series on common dates → x[], y[] (n pairs)
r ← Σ(xᵢ−x̄)(yᵢ−ȳ) / √(Σ(xᵢ−x̄)² · Σ(yᵢ−ȳ)²)
t² ← r²(n−2)/(1−r²)
p ← I_{(n−2)/((n−2)+t²)}((n−2)/2, 1/2) # regularised incomplete beta (two-tailed)
report r, R² = r², p; warn if n < 8, frequencies differ,
or |r_detrended| (on month-to-month changes) ≪ |r|
```
OLS trend and illustrative forecast:

```
slope ← Σ(i−ī)(vᵢ−v̄) / Σ(i−ī)² over index i = 0…n−1
intercept ← v̄ − slope·ī
trend line ← slope·i + intercept; extrapolate k periods, labelled "illustrative only"
```
3.7.6 Input and output design. *Inputs:* date-range pickers (From/To) on every data page; range-preset chips (1Y/3Y/5Y/All); grouped indicator selectors driven by the live catalogue (single-select for profiles, multi-select for briefings); a currency-converter amount field; CSV upload/paste in the Playground; URL query parameters (?from=&to=&view=, ?ids=) so any composed view is shareable and regenerable. All inputs are validated client-side and, for the API, server-side. *Outputs:* interactive charts with plain-language captions; statistic tiles; sortable data tables; CSV downloads; attributed PNG chart exports; APA citations; the print-ready briefing document; and machine outputs (JSON with _links, the llms.txt discovery file).

3.7.7 Security and data-integrity design. Because the system is public-read by design, the relevant security question is one of integrity rather than secrecy, ensuring that nothing is corrupted or duplicated, rather than concealing anything. The browser-side database credential is a public anonymous key gated by PostgreSQL row-level security, and is therefore read-only by construction; it cannot modify data, which is what makes shipping it client-side safe rather than reckless. Every write path is demo-safe by default (ALLOW_DATA_WRITES=false), and the public validation endpoint is structurally incapable of writing at all, not merely configured not to. A uniqueness constraint on (indicator_id, obs_date) prevents duplicate ingestion at the database level regardless of the application code. Any user-supplied content echoed back by the frontend, such as Playground verdicts, is HTML-escaped, and adversarial inputs, script tags, NaN/Infinity, and a 5, 000-row flood, were added to the test suite specifically to confirm that those paths hold. Real secrets, namely the service keys, exist only as server-side environment variables and never enter the repository.

3.7.8 User-interface design. The dashboard uses a dark "Lagos Noir" theme with the same storytelling pattern repeated on every indicator page, a headline statistic, three short blocks covering what happened, why it matters and how to read the chart, the chart itself, and a data table underneath, plus a "markets-terminal" treatment on the charts themselves: range selectors, a hover crosshair, end-of-line value tags, event markers. Two things here are treated as requirements rather than aesthetic preference. The first is honest encoding: no dual y-axes anywhere (differing scales become aligned panels or z-scores instead), no data-censoring transforms, and units always stated plainly. The second is layered depth, the Reader/Analyst dial allows the general public and a researcher to use the same page rather than forking the interface into two separate products for two separate audiences. Accessibility (WCAG 2.1 AA contrast, keyboard focus, aria labelling and reduced-motion support) was maintained as a constraint throughout, rather than addressed in a final review.

CHAPTER FOUR, SYSTEM IMPLEMENTATION AND TESTING

4.1 Introduction

Chapter Three set out what was to be built. This chapter is the account of building it, the tools used, how the seven-stage pipeline came together in practice, the points at which implementation forced a decision the design had not anticipated, and how the result was tested and evaluated.

4.2 Development Tools and Technologies

The backend was implemented in Python 3 using the FastAPI framework, served by the Uvicorn ASGI server, with supabase-py for database access. Data resides in Supabase (managed PostgreSQL), accessed under row-level security. The frontend was built with HTML5, CSS3 and vanilla JavaScript, using Chart.js v4 together with its annotation and zoom plugins for visualisation. Version control, continuous integration and deployment were handled through GitHub and GitHub Actions, with the dashboard hosted on GitHub Pages and the API on Render. Testing employed Pytest, browser-based functional testing, and Lighthouse accessibility audits.

4.3 Implementation of the Pipeline

The pipeline was implemented stage by stage. Collection obtained source data from CBN, NBS and World Bank publications, arranged into CSV and Excel inputs that loader scripts (project/etl/, project/database/seed/) read into the database, with a reproducible seed snapshot (observations.csv, approximately 12, 100 rows) allowing the whole dataset to be recreated. Validation checks each row's types and dates and normalises fields before any write, rejecting invalid rows with clear errors. Standardisation reduces all series to the common (indicator_id, obs_date, value) observation form, with indicator metadata (unit, source and frequency) held separately. Storage places the data in three PostgreSQL tables on Supabase, where the anonymous key is public and read-only under row-level security. The analytics layer computes the latest value, period and year-on-year change, trend direction, a simple ordinary-least-squares forecast, and Pearson correlation (for example, inflation against the exchange rate), with the compare and analytics pages replicating the correlation client-side in JavaScript. The API is provided by FastAPI, which exposes the versioned endpoints, auto-generates OpenAPI/Swagger documentation at /docs, and adds HATEOAS _links to every response, running with proxy headers in production so that hypermedia links honour HTTPS. Finally, presentation is handled by the static dashboard, which fetches data directly from Supabase for the charts and renders it with Chart.js, applying the storytelling and terminal-chart patterns.

4.4 Key Implementation Highlights

A few decisions from the build warrant separate attention, since they became apparent only on encountering the problems they solve.

The whole API consolidates into one canonical main.py rather than being scattered across files that drift apart, with credentials read from the environment and safe fallbacks if they are missing. Unit correctness became a first-class concern rather than a detail, because raw balance-sheet values are stored in naira thousands while financial-statement values are stored in naira millions, mixing those up anywhere in the UI would silently misstate a figure by orders of magnitude, so both get converted and clearly labelled (naira trillions, for instance) before they ever reach a chart. And the charting logic itself (shared.js) is one shared, reusable set of utilities, the takeaway stat, the range selector, the crosshair, the end-of-line labels, used the same way on every page, rather than each page reinventing its own version.

4.5 Analytical Methods and Their Limitations

Every analytic on the platform is computed transparently in the browser, directly from the aggregated data, across the full catalogue of indicators. The methods are deliberately classical and explainable rather than resembling a black box, so that any result a user sees can be reproduced and checked by hand. This reflects a guiding principle of the project: correctness and honesty are valued above impressiveness, and where a result is unreliable, the platform states as much rather than concealing it.

Several analytical methods are implemented. Descriptive statistics cover the latest value, period and year-on-year change, the minimum and maximum with their dates, the mean, the standard deviation as a volatility measure, and the coefficient of variation. Trend estimation uses Ordinary Least Squares (OLS) linear regression, reporting the slope, the coefficient of determination (R^2), and the correlation of the series with time. Correlation analysis computes the Pearson product-moment correlation coefficient on two series' date-aligned observations, reported together with R^2 (the share of variance explained) and a two-tailed statistical-significance p-value derived from the Student-t distribution via the regularised incomplete beta function. Standardisation applies z-score normalisation, standard deviations from each series' own mean, so that two indicators measured in different units can be compared on a single, honest axis rather than a misleading dual axis. A trend-robustness, or spurious-correlation, check compares the detrended first-difference correlation with the level correlation in order to flag relationships driven mainly by a shared trend rather than a direct association. Forecasting extrapolates the OLS trend linearly a few periods ahead and is presented explicitly as illustrative only. Finally, reliability guards issue automatic warnings for correlations computed over very few overlapping observations or between series of differing reporting frequency.

The analytical layer is subject to several acknowledged limitations. The figures are not real-time, since they represent a manually ingested snapshot rather than an automated live pipeline. No seasonal adjustment is applied; monthly and quarterly series are presented as reported. The forecast is a straight-line OLS extrapolation without a confidence or prediction interval, and no ARIMA, exponential-smoothing or other time-series model is used. The platform measures association rather than causation: it computes correlation only, and performs no causality testing (such as Granger causality) or lead/lag cross-correlation analysis. Short or mixed-frequency comparisons are weaker, and although flagged to the user they remain available. Coverage is bounded by the sources, so that some series end earlier than others, the annual financial statement, for example, ends in 2012. These limitations are presented deliberately, on the principle that a smaller, correct and trustworthy platform is preferable to a larger one that overstates its capabilities, and each is a candidate for the future work discussed in Chapter Five.

4.6 System Screenshots

The figures below are actual captures of the deployed system.

Figure 4.1, Homepage: live KPI cards with trend sparklines and coverage strip

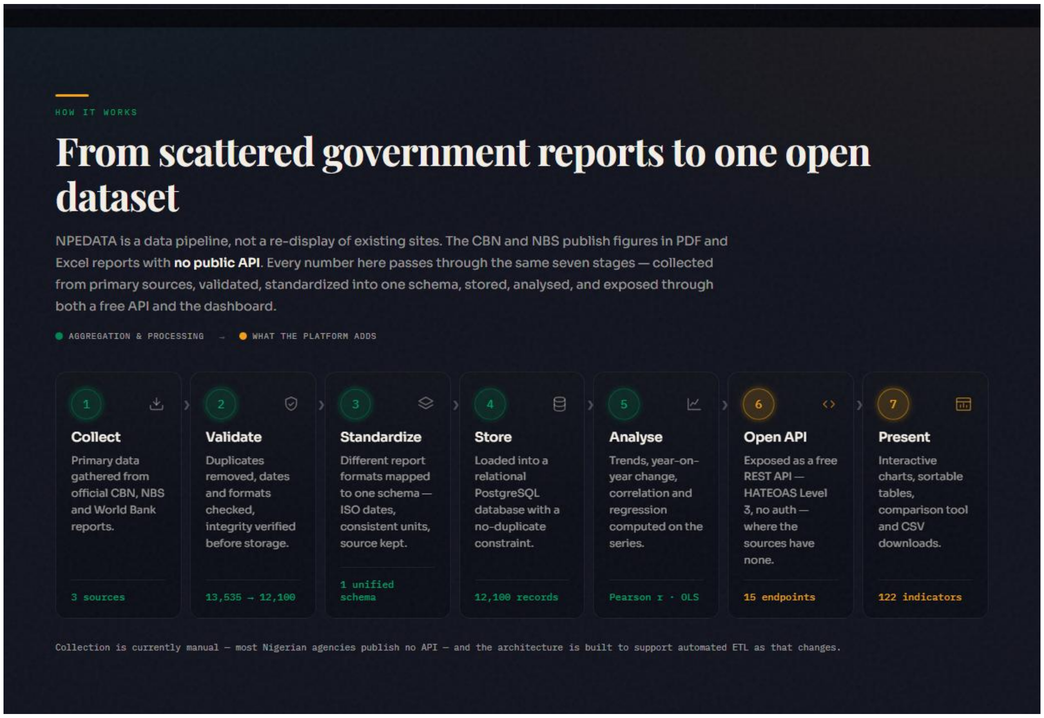


Figure 4.2, The seven-stage "How It Works" pipeline section

Figure 4.3, Flagship story: inflation and the NGN/USD rate as aligned single-axis panels with event markers

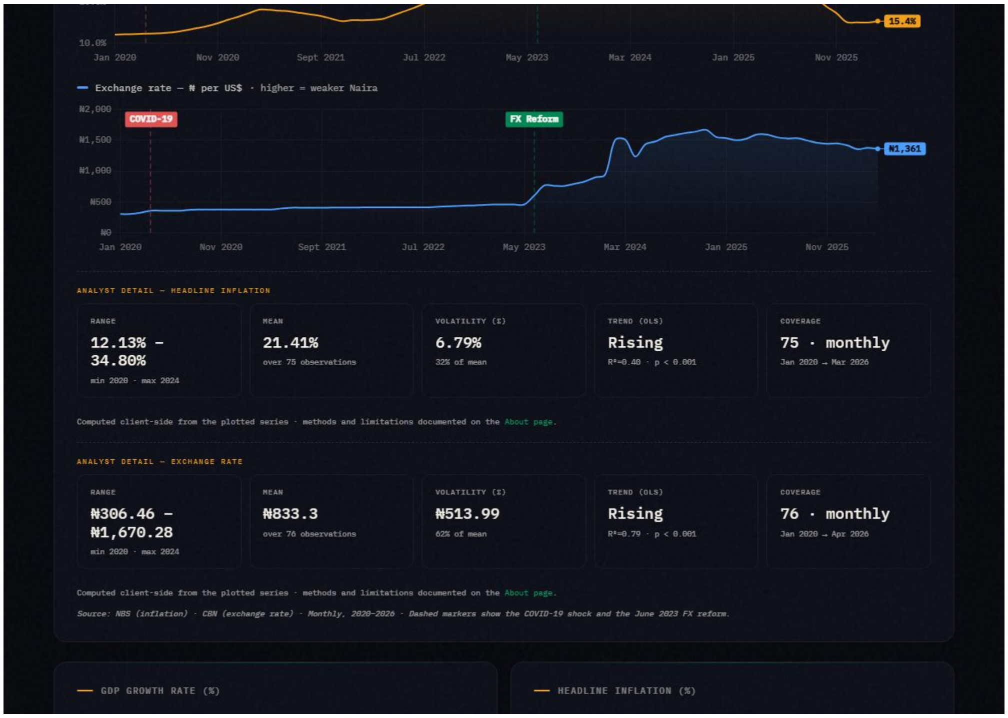


Figure 4.4, The Reader/Analyst dial revealing statistical panels on the homepage

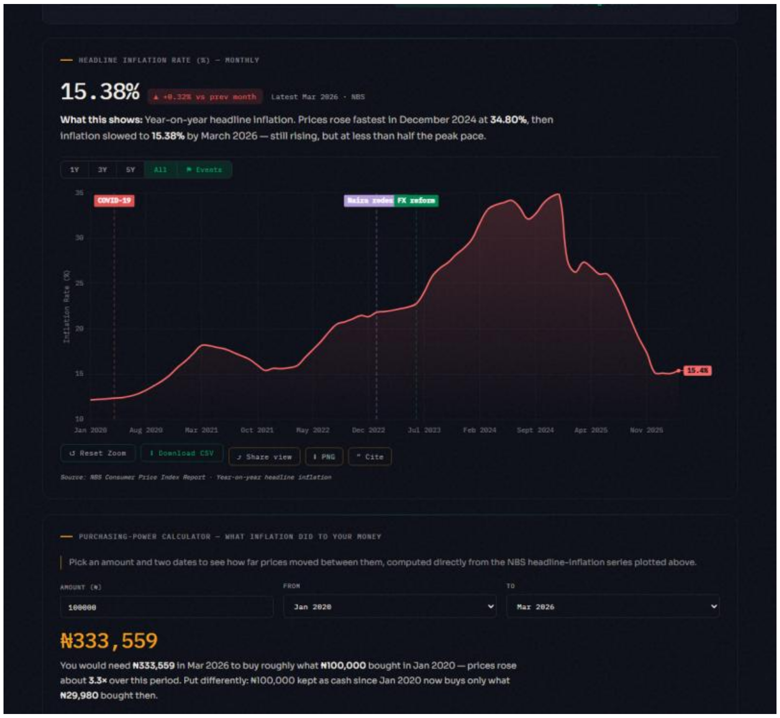


Figure 4.5, Inflation page: research toolkit (Share/PNG/Cite), event toggle and purchasing-power calculator

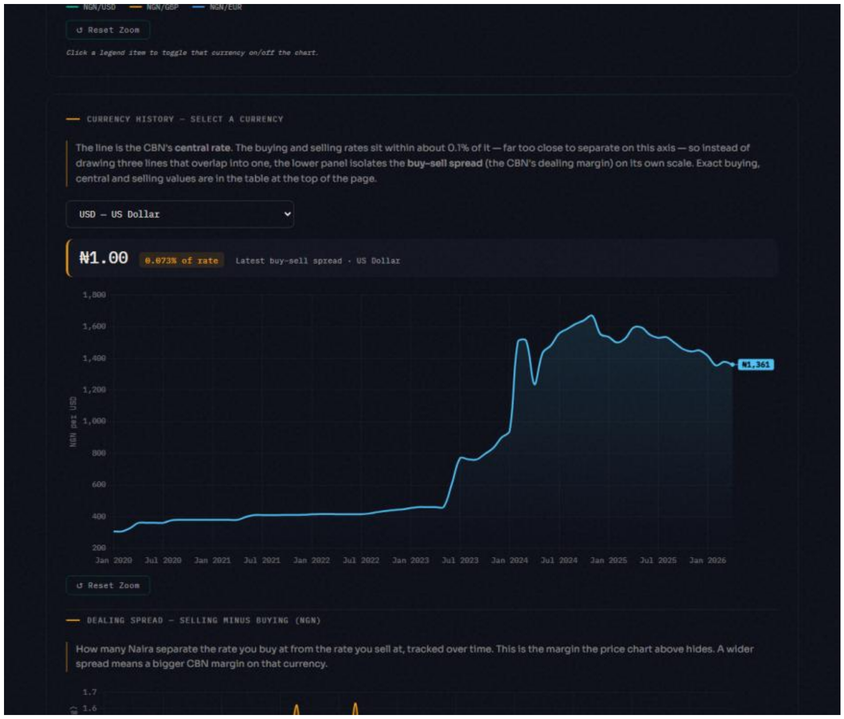


Figure 4.6, Multi-currency page: central rate with the isolated dealing-spread panel

Figure 4.7, CBN balance sheet: gold, government and bankers' deposits as small multiples

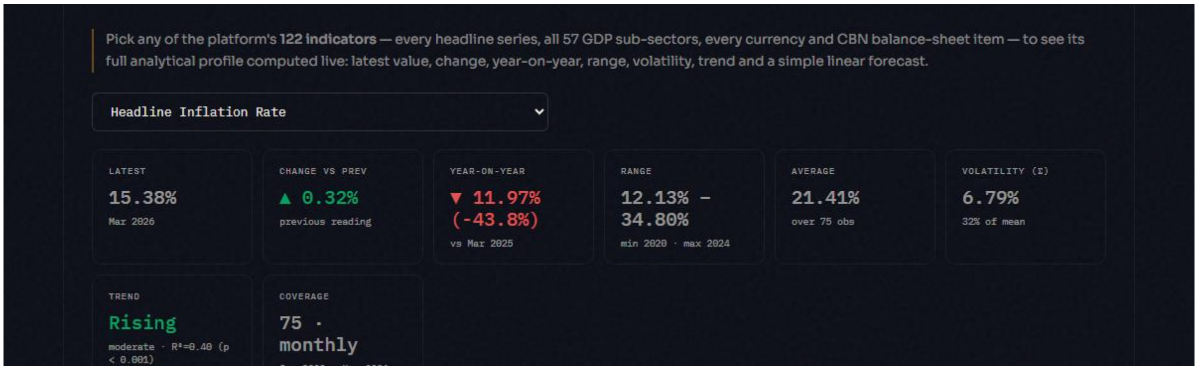


Figure 4.8, "Analyze Any Indicator": statistical profile tiles with trend R² and p-value

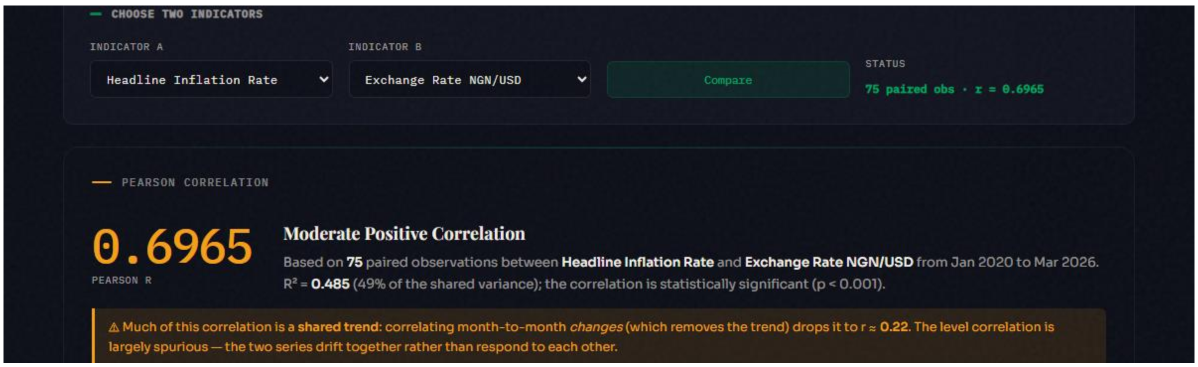


Figure 4.9, Compare tool reporting Pearson r with R² and statistical significance

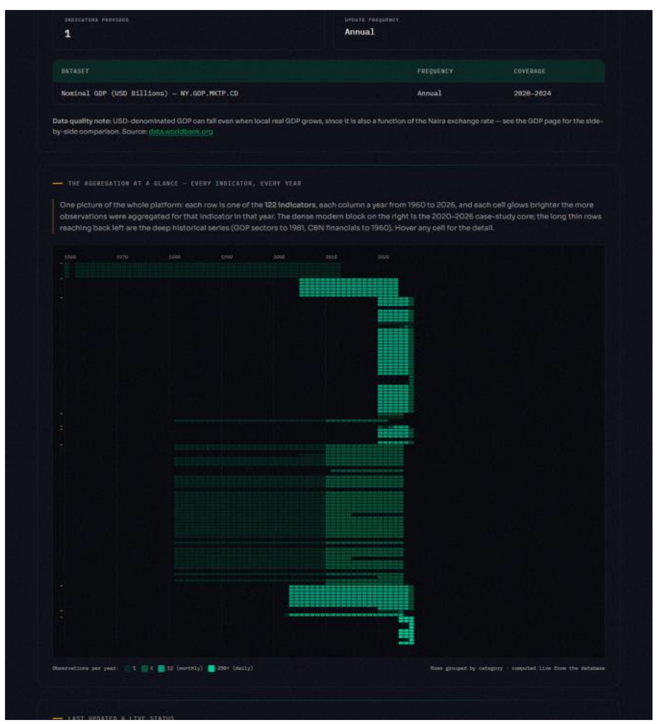


Figure 4.10, The Aggregation Heatmap: 122 indicators × 1960–2026, computed live

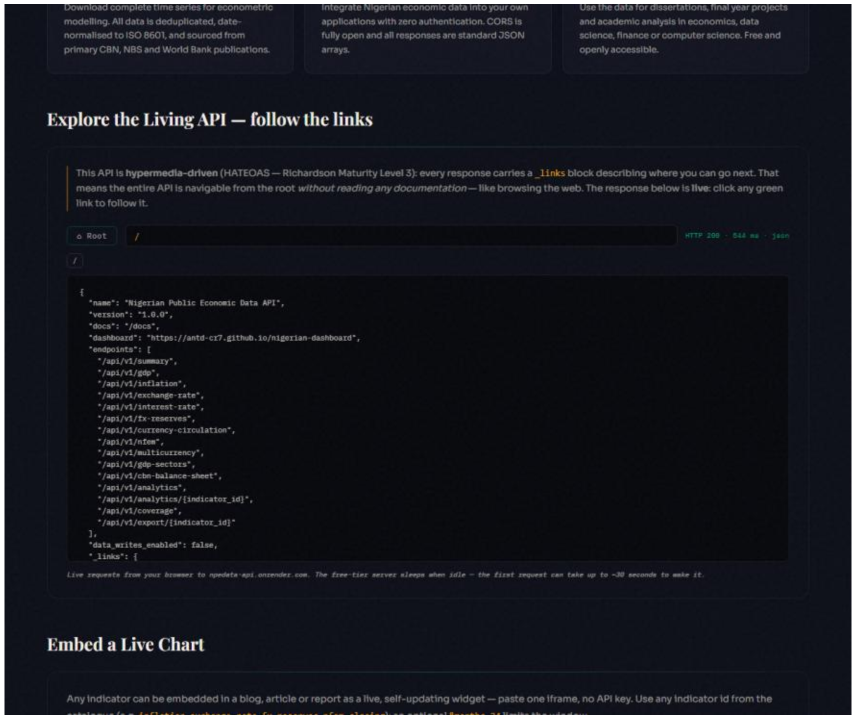


Figure 4.11, HATEOAS Explorer browsing the live API by following _links

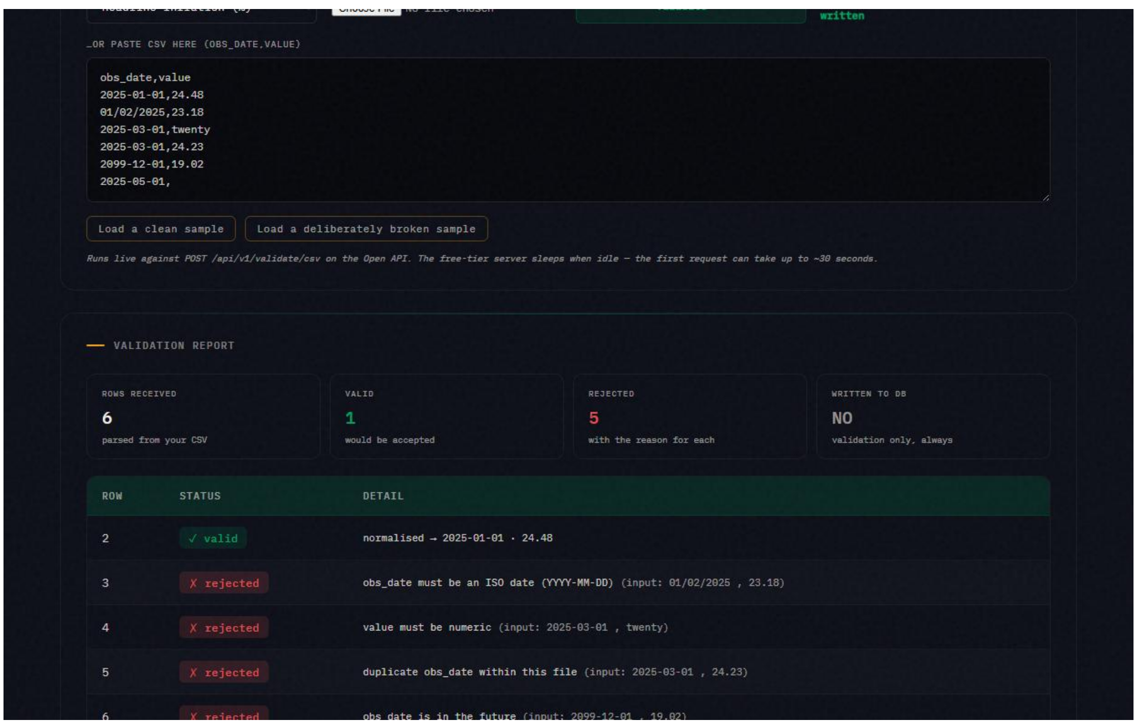
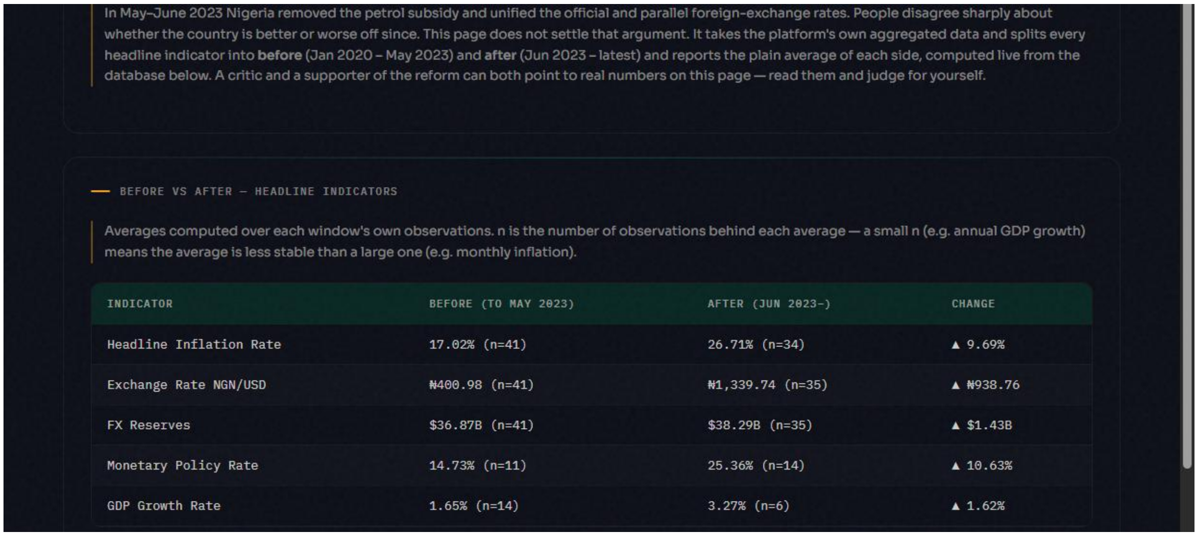


Figure 4.12, Pipeline Playground: per-row validation verdicts, nothing written

Figure 4.13, Briefing Studio: generated, cited, print-ready economic briefing

Figure 4.14, Correlation Matrix: 12 headline indicators cross-correlated on annual averages, computed live



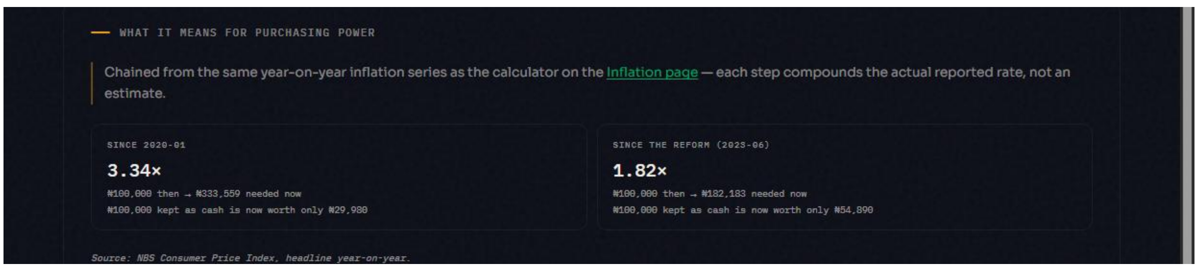
In May–June 2023 Nigeria removed the petrol subsidy and unified the official and parallel foreign-exchange rates. People disagree sharply about whether the country is better or worse off since. This page does not settle that argument. It takes the platform's own aggregated data and splits every headline indicator into **before** (Jan 2020 – May 2023) and **after** (Jun 2023 – latest) and reports the plain average of each side, computed live from the database below. A critic and a supporter of the reform can both point to real numbers on this page — read them and judge for yourself.

BEFORE VS AFTER — HEADLINE INDICATORS

Averages computed over each window's own observations. n is the number of observations behind each average — a small n (e.g. annual GDP growth) means the average is less stable than a large one (e.g. monthly inflation).

INDICATOR	BEFORE (TO MAY 2023)	AFTER (JUN 2023–)	CHANGE
Headline Inflation Rate	17.02% (n=41)	26.71% (n=34)	▲ 9.69%
Exchange Rate NGN/USD	₦400.98 (n=41)	₦1,339.74 (n=35)	▲ ₦938.76
FX Reserves	$36.87B (n=41)	$38.29B (n=35)	▲ $1.43B
Monetary Policy Rate	14.73% (n=11)	25.36% (n=14)	▲ 10.63%
GDP Growth Rate	1.65% (n=14)	3.27% (n=6)	▲ 1.62%

Figure 4.15, Reform Impact: before/after June 2023 averages for five headline indicators, with neutral critic/supporter readings generated from the same numbers



WHAT IT MEANS FOR PURCHASING POWER

Chained from the same year-on-year inflation series as the calculator on the Inflation page — each step compounds the actual reported rate, not an estimate.

SINCE 2020-01

3.34×

₦100,000 then → ₦333,559 needed now

₦100,000 kept as cash is now worth only ₦29,980

SINCE THE REFORM (2023-06)

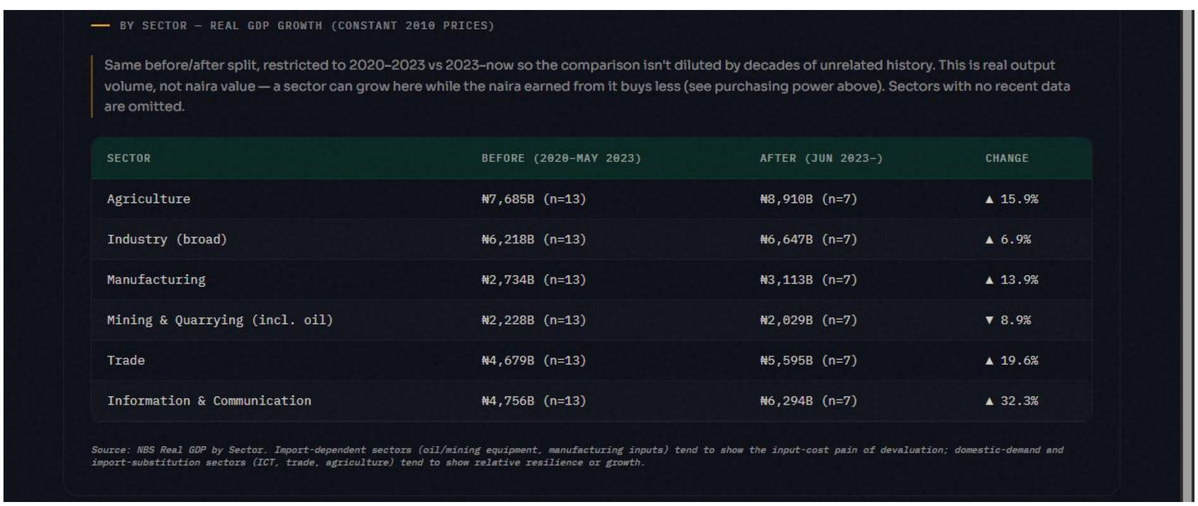
1.82×

₦100,000 then → ₦182,183 needed now

₦100,000 kept as cash is now worth only ₦54,890

Source: NBS Consumer Price Index, headline year-on-year.

Figure 4.16, Reform Impact: purchasing-power erosion since January 2020 and since the reform, chained live from the same year-on-year inflation series used by the Inflation page's calculator



BY SECTOR — REAL GDP GROWTH (CONSTANT 2010 PRICES)

Same before/after split, restricted to 2020–2023 vs 2023–now so the comparison isn't diluted by decades of unrelated history. This is real output volume, not naira value — a sector can grow here while the naira earned from it buys less (see purchasing power above). Sectors with no recent data are omitted.

SECTOR	BEFORE (2020–MAY 2023)	AFTER (JUN 2023–)	CHANGE
Agriculture	₦7,685B (n=13)	₦8,910B (n=7)	▲ 15.9%
Industry (broad)	₦6,218B (n=13)	₦6,647B (n=7)	▲ 6.9%
Manufacturing	₦2,734B (n=13)	₦3,113B (n=7)	▲ 13.9%
Mining & Quarrying (incl. oil)	₦2,228B (n=13)	₦2,029B (n=7)	▼ 8.9%
Trade	₦4,679B (n=13)	₦5,595B (n=7)	▲ 19.6%
Information & Communication	₦4,756B (n=13)	₦6,294B (n=7)	▲ 32.3%

Source: NBS Real GDP by Sector. Import-dependent sectors (oil/mining equipment, manufacturing inputs) tend to show the input-cost pain of devaluation; domestic-demand and import-substitution sectors (ICT, trade, agriculture) tend to show relative resilience or growth.

Figure 4.17, Reform Impact: real GDP growth by sector, before/after June 2023, computed live and windowed to 2020–2026 to avoid diluting the comparison with unrelated history

The Reform Impact page also carries a "Who this affects, and why" section covering global and institutional investors, entrepreneurs and MSMEs, domestic Nigerian investors, and why foreign direct investment might follow the reform, explicitly labelled as economic reasoning applied to the numbers above rather than a further live computation, so the distinction between what the platform computed and what it is arguing stays honest.

4.7 Testing and Validation

I validated the platform through several complementary strands rather than relying on any one of them alone, automated tests, independent statistical validation, and a systematic audit of the data itself.

Automated unit testing (Pytest). The Open API is covered by a suite of **24 unit tests** in tests/test_main.py, exercising the read endpoints, the demo-safe ingestion path, the TTL cache (round-trip, expiry and size-cap eviction), and, importantly, asserting the presence and correctness of the HATEOAS _links blocks that make the API Level 3. All 24 tests pass.

Statistical validation. Because the analytics report inferential statistics, the underlying mathematics was validated independently of the user interface. The correlation-significance function, a two-tailed Student-t p-value computed via the regularised incomplete beta function, was checked against known reference cases; for example, $r = 0.70$ over $n = 75$ yields $p \approx 4 \times 10^{-12}$, while $r = 0.20$ over $n = 20$ yields $p \approx 0.40$, which is correctly *not* significant. The scale-aware value formatter was unit-tested across every unit type in the catalogue (percentages, exchange rates, USD billions, and the naira thousands, millions and billions scales), confirming, for instance, that CBN total assets render as ₦81.04T rather than as a raw or mislabelled figure. The analytics engine was also exercised against deliberately awkward real series, a daily series (NFEM), a series containing negative values (government deposits), a sparse four-point series

(AED rates), a count series, and a series whose coverage ends early (services GDP), confirming correct output and no crashes on these edge cases.

Data-truthfulness auditing. Every chart's stated figures, ranges and units were cross-checked against the stored data, and this is the strand that surfaced genuine problems rather than merely confirming that matters were already sound. The audit identified a data-censoring routine that had been silently capping some balance-sheet series, hiding the gold revaluation and understating bankers' deposits; unit mislabels that mis-scaled monetary values by a factor of a thousand; a chart titled an "inverse relationship" that the data in fact showed to be a weak positive one ($r \approx 0.33$), the opposite of its own caption; and a sector comparison that had inadvertently placed figures from two different years side by side as though they were comparable. Each defect was traced back to the original source figures and corrected, and the process is documented here rather than presenting the first version as though it had been correct throughout.

Functional and visual verification. Every dashboard page was loaded against the live database and inspected, including through automated screenshots, to confirm that charts, date filters, indicator comparisons, table sorting and CSV downloads behave correctly, and that the responsive layout holds across screen widths.

Accessibility testing. Colour contrast and related criteria were checked against the WCAG 2.1 AA standard (contrast ratio ≥ 4.5:1) using Lighthouse audits; contrast defects found during development were corrected.

API testing. Endpoints were exercised manually through the browser, the auto-generated Swagger UI at /docs, and curl, confirming correct payloads, CSV export with the RFC 8288 Link header, and the demo-safe behaviour of the ingestion endpoints.

Table 4.1, Representative test cases

#	Test	Expected	Result
1	GET /api/v1/summary	5 headline indicators + _links	Pass
2	GET /api/v1/analytics/inflation	latest, change, trend, forecast	Pass
3	GET /api/v1/export/exchange_rate	CSV + RFC 8288 Link header	Pass
4	POST /api/v1/ingest/csv (demo mode)	validated, not written to DB	Pass
5	Full Pytest suite	24 / 24 tests pass	Pass
6	Correlation p-value vs known cases	matches reference values	Pass
7	Value formatter across all unit types	correct scale and symbol	Pass
8	Analytics engine on edge-case series	no crash, correct output	Pass
9	Chart figures vs stored data	exact match, correct units	Pass (after fixes)
10	Accessibility contrast	≥ 4.5:1 (WCAG 2.1 AA)	Pass

4.8 Results and Discussion

The result is a platform that aggregates 122 indicators and approximately 12, 100 observations into one queryable store, serves them through a documented open API with hypermedia controls, and presents them through a dashboard that is accessible in practice rather than only in principle. Testing confirmed this: the API behaves as specified, and the visualisations proved to be both correct and clearly explained once the unit-correctness audit had been completed. That audit is arguably the most significant part of Chapter Four, since it marks the difference between a platform that merely appears trustworthy and one that is.

4.9 Technology-Stack Assessment

Once the build was largely complete, the stack was re-evaluated against a single question: is any of these technology choices limiting the platform? To keep the assessment evidence-based rather than a matter of taste, it draws on the measurements gathered during stress testing rather than on opinion.

Two choices proved themselves and were kept deliberately. The static HTML, CSS and vanilla JavaScript frontend, with no framework and no build step, delivered a 100/100 Lighthouse accessibility score at this scale, together with zero build or dependency maintenance, free hosting, and complete auditability, since the entire implementation can be read directly from source. The known weakness of the approach, namely duplicated code drifting apart across pages, was addressed architecturally with a single shared library (shared.js) carrying all chart, analytics and UI logic, so that a fix in one place applies everywhere. The FastAPI and Supabase (PostgreSQL) combination likewise handled every stress-test scenario, including a 5, 000-row adversarial CSV, with all 17 live endpoints passing, while the relational tidy-observations model absorbed daily, monthly, quarterly and annual series in a single schema.

Three limitations were measured, and each was addressed. First, the free-tier API exhibits cold starts of around 30 seconds, because Render's free tier sleeps after idle; this was mitigated in three ways, a scheduled keep-alive workflow pings the service every ten minutes, every API-dependent page shows an explicit "server waking" state instead of appearing broken, and the dashboard is architecturally independent of the API (reading the database's REST layer directly), so that the primary user experience cannot be taken down by the API host. Second, repeated-query latency on the multi-currency endpoint (33 series) reached about 5.4 seconds per request, because every call repeated 33 database round-trips; a size-capped, five-minute-TTL in-process cache was added at the data-access layer, appropriate because the dataset changes only at ingestion time, cutting repeat responses to well under a second and reducing database load, which is the standard first scaling step of caching applied within the existing stack rather than replacing it. Third, PostgREST imposes row-window limits, returning at most around 1, 000 rows per request, so the coverage heatmap fetches the full 12, 100-observation dataset through paged parallel requests; this is acceptable at the present scale, and a server-side aggregate endpoint is the documented next step should the dataset grow.

Scaling path (if requirements outgrow the stack): the layers are decoupled, so each can be upgraded independently without a rewrite, the database is already PostgreSQL (scales to a paid tier unchanged); the API is containerisable FastAPI (moves to an always-on host, gaining zero-cold-start); and because all data access goes through the documented Open API or the database's REST layer, the frontend could be rebuilt in any framework without touching the backend. The conclusion of the assessment is that the stack is not the limiting factor at the platform's current scope; where friction was measured, it was engineered around within the stack, and the upgrade path for each layer is documented.

CHAPTER FIVE, SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary

This project set out to address the fragmentation and inaccessibility of public Nigerian economic data, and the result is NPEDATA, a seven-stage platform that collects, validates, standardises, stores, analyses and publishes economic indicators through both a dashboard and a free open API. Measured against the objectives in Chapter One, all six were substantively met rather than nominally satisfied: the data is aggregated into one unified model, the analytics are implemented and functioning, the API is HATEOAS-compliant and documented, the dashboard is accessible and explanatory rather than merely decorative, and the system has been tested for both correctness and accessibility rather than assumed to be sound.

Table 5.1, Achievement of objectives

#	Objective (Ch. 1)	Delivered evidence
1	Collect indicators from CBN, NBS, World Bank	122 indicators, ~12, 100 observations; reproducible seed snapshot
2	Unified standardised data model	One tidy observations schema holding daily-to-annual series; data dictionary Section 3.7.3
3	Server-side analytics	Descriptives, YoY, OLS trend, correlation with R^2/p-value; TTL-cached API
4	Free documented HATEOAS API	Versioned endpoints, _links throughout, RFC 8288 on CSV; interactive Explorer
5	Clear, truthful dashboard	Storytelling pattern, single-axis policy, Reader/Analyst dial, WCAG AA 100/100
6	Test and evaluate	24-test suite, statistical validation, adversarial stress test, live sweeps (Section 4.7)

5.2 Conclusion

The central demonstration of this project is that Nigeria's scattered, non-machine-readable public economic data need not remain in that state: it can be consolidated into a resource that is correct and programmatically usable using only free and open-source tools, without a government contract or a commercial budget. In doing so, it lowers the barrier to using this data for students, researchers, developers and the public, and offers a reasonably candid reference for what open-data practice could look like in a Nigerian context, including the limitations documented rather than concealed in Chapter Four.

5.3 Recommendations and Future Work

The project is not finished, and an undertaking of this scope should not present itself as such. Several directions are recommended for future work, whether by the author or by others. The most immediate is to automate collection, using scheduled scrapers or connectors to the source portals, so that freshness no longer depends on manual re-ingestion. Coverage could then be expanded to include more indicators, state-level rather than only national data, and longer historical series where the sources allow. Authentication tiers and rate limiting could be introduced once there are API consumers heavy enough to require them. Richer analytics, such as seasonal adjustment or additional forecasting methods, could be added without sacrificing the transparency of the current classical approach. Client software development kits, a Python or JavaScript package, would lower the remaining friction in adopting the API. Finally, the data-quality workflow itself could be formalised, with automated validation dashboards and clearer provenance tracking, rather than relying on the kind of manual audit described in Section 4.7.

None of these recommendations addresses the deeper barrier discussed in Section 2.4, the institutional reluctance to share data that has kept the CBN, NBS and other agencies from undertaking this themselves. A student project cannot legislate that away, and nothing in this report resolves it. If, however, a reference implementation of this kind is used, cited, or simply noticed, it becomes one small piece of evidence for the argument that BudgIT and others already advance: that unifying this data was always a governance choice rather than an engineering problem (Anintah, n.d.). That, more than any single feature, is the contribution this project would ultimately hope to make.

5.4 Contribution to Knowledge

The concrete contribution of this project is a working, reproducible, openly licensed reference implementation of a unified Nigerian public-economic-data platform with a fully HATEOAS-level open API, an artefact that, as far as the review in Chapter Two could establish, did not previously exist in freely accessible form, together with a demonstration that such a platform is achievable using entirely free tooling, by a single student, within one academic year.

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APPENDICES

- **Appendix A, Sample API response** (showing the _links HATEOAS block).
- **Appendix B, Selected source code** (data model, an endpoint, the analytics function).
- **Appendix C, Full list of endpoints** (see project README.md).
- **Appendix D, Screenshots** (full set referenced in Chapter Four).
- **Appendix E, Repository & live links:**
- Source: https://github.com/ANTD-CR7/nigerian-dashboard
- Dashboard: https://antd-cr7.github.io/nigerian-dashboard/
- API: https://npedata-api.onrender.com (docs at /docs)